

Shiv Nadar University Chennai
Department of Computer Science and Engineering
CS3807 – Deep Learning Laboratory
Laboratory Report – Experiment 7
End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising
Autoencoders, and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester: V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27
Student Name / Reg.	Lab Student / 2024-AI-01	Date: 30-09-2026

1 Objective and Theoretical Overview

The objective of this laboratory study is to develop a comprehensive, end-to-end understanding of representation learning, dimensionality reduction, noise suppression, and deep generative modeling across four distinct autoencoder paradigms:

- Fully Connected Autoencoder (FC-AE):** Compressing vectorized 784-dimensional images into a deterministic 16-dimensional bottleneck vector using dense transformations.
- Convolutional Autoencoder (CAE):** Preserving spatial locality, parameter sharing, and translation equivariance using 2D convolutions, max-pooling, and upsampling operators.
- Denoising Autoencoder (DAE):** Compelling the latent manifold to capture robust structural invariants by reconstructing clean ground-truth targets from corrupted inputs.
- Variational Autoencoder (VAE):** Learning an intractable continuous probabilistic latent manifold parameterized by posterior parameters $\mu(x)$ and $\log \sigma^2(x)$, regularized via Kullback-Leibler (KL) divergence, and sampled using the reparameterization trick.

2 Dataset and Experimental Setup

All experiments are executed on the benchmark MNIST handwritten digit dataset ($28 \times 28 \times 1$ grayscale images, classes 0 to 9). A controlled protocol is adhered to throughout the pipeline:

- **Training Partition:** Exactly 10,000 images normalized to the floating-point range $[0.0, 1.0]$.
- **Test/Validation Partition:** Exactly 2,000 unseen images normalized to $[0.0, 1.0]$.
- **Tensors:** Flattened matrices of shape $(N, 784)$ for the FC-AE, and rank-4 image tensors of shape $(N, 28, 28, 1)$ for CAE, DAE, and VAE.
- **Hyperparameters:** Adam optimizer with learning rate $\eta = 10^{-3}$, mini-batch size $B = 128$, training duration of 20 epochs, and Binary Cross-Entropy (BCE) reconstruction loss.

3 Source Code and Implementation Repository

The complete modular PyTorch codebase, reproduction scripts, generated visualization plots, and evaluation logs are hosted at the following repository:

GitHub Repository: <https://github.com/Kamlesh-DevOP/deep-learning-lab/tree/main/lab-7>

4 Model Architectures and Mathematical Formulations

4.1 Deterministic Autoencoders (FC-AE, CAE, DAE)

Deterministic autoencoders train an encoder f_θ and decoder g_ϕ to minimize the reconstruction discrepancy between input $x \in [0, 1]^D$ and reconstruction $\hat{x} = g_\phi(f_\theta(x))$:

$$\mathcal{L}_{\text{BCE}}(x, \hat{x}) = -\frac{1}{D} \sum_{j=1}^D [x_j \log \hat{x}_j + (1 - x_j) \log(1 - \hat{x}_j)] \quad (1)$$

In DAE training, the encoder receives corrupted inputs $\tilde{x} = \text{clip}(x + \mathcal{N}(0, \sigma^2), 0, 1)$, while the reconstruction target remains the original clean image x .

4.2 Variational Autoencoder (VAE) Formulation

The VAE optimizes the Evidence Lower Bound (ELBO) consisting of reconstruction fidelity and relative entropy with respect to an isotropic prior $p(z) = \mathcal{N}(0, I)$:

$$\mathcal{L}_{\text{VAE}}(\theta, \phi; x) = \mathcal{L}_{\text{rec}}(x, \hat{x}) + D_{\text{KL}}(q_\phi(z|x) \parallel p(z)) \quad (2)$$

For latent dimension $d_z = 2$, the analytical Kullback-Leibler divergence is computed as:

$$D_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^{d_z} (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2) \quad (3)$$

Stochastic sampling is rendered differentiable via the reparameterization trick: $z = \mu(x) + \sigma(x) \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$.

5 Empirical Results, Visualizations, and Inferences

5.1 Study 1: Fully Connected Autoencoder

The FC-AE maps 784 input dimensions through Dense(128) $\rightarrow$ Dense(32) $\rightarrow$ Dense(16) bottleneck, reconstructed via Dense(32) $\rightarrow$ Dense(128) $\rightarrow$ Dense(784).

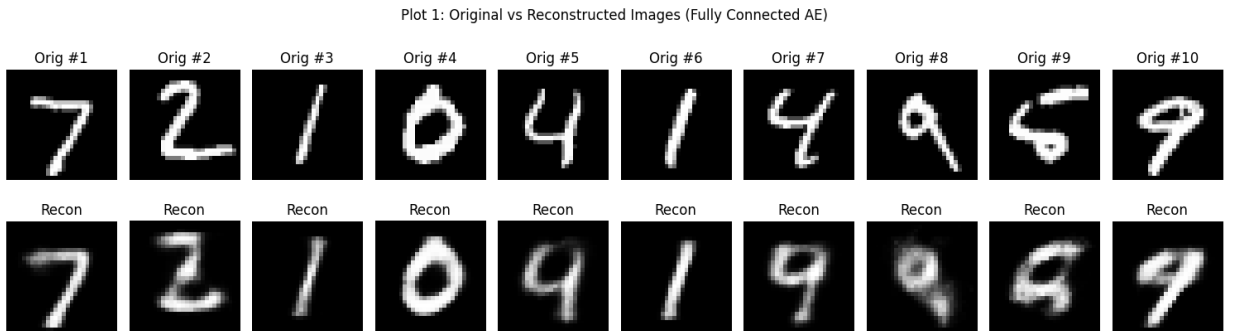


Figure 1: Original vs. Reconstructed Test Digits (Fully Connected Autoencoder).

Inference for Plot 1: This figure presents a qualitative comparison between ten original test digit samples and their reconstructions produced by the fully connected autoencoder. While the gross structural morphology of each digit is preserved (e.g., vertical strokes of ‘1’ and broad loops of ‘0’), fine stroke details exhibit noticeable blurriness, and closed contours in complex digits such as ‘8’ and ‘9’ suffer from contrast attenuation. This degradation occurs because vectorizing 28×28 images into 784-dimensional arrays destroys 2D spatial adjacency, forcing dense affine layers to compress all pixel interactions into an undercomplete 16-dimensional bottleneck under a pixel-wise loss that penalizes high-frequency variance. We conclude that while fully connected autoencoders succeed at coarse topological compression, they inherently struggle with sharp boundary reconstruction due to the absence of geometric inductive bias.

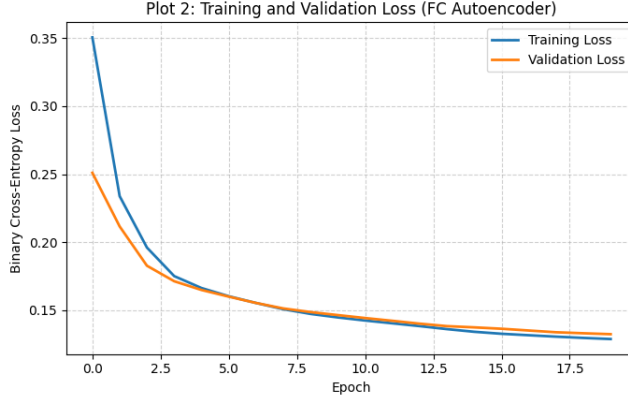


Figure 2: Training and Validation Loss Trajectories (FC Autoencoder).

Inference for Plot 2: This plot tracks the binary cross-entropy reconstruction loss for both training and validation sets across 20 training epochs. A rapid decay occurs over the first five epochs (loss dropping from 0.3478 down to 0.1709), transitioning smoothly into asymptotic stabilization, terminating at a training loss of 0.1288 and validation loss of 0.1325. The close alignment between the two curves without divergence indicates that the Adam optimizer navigates the parameter space smoothly, and the compact 16-dimensional bottleneck imposes effective structural regularization against overfitting. We conclude that the FC-AE architecture converges stably and achieves robust generalization without memorizing dataset-specific noise.

Table 1: Reconstruction Performance Metrics of the Basic Fully Connected Autoencoder

Metric	Experimental Value
Test Mean Squared Error (MSE)	0.02389
Test Mean Absolute Error (MAE)	0.06188
Mean Structural Similarity Index (SSIM)	0.7193
Total Trainable Parameters	211,040
Total Training Execution Time (20 Epochs)	23.31 s
Final Training / Validation Loss (BCE)	0.1288 / 0.1325

5.2 Study 2: Convolutional Autoencoder (CAE) vs. FC-AE

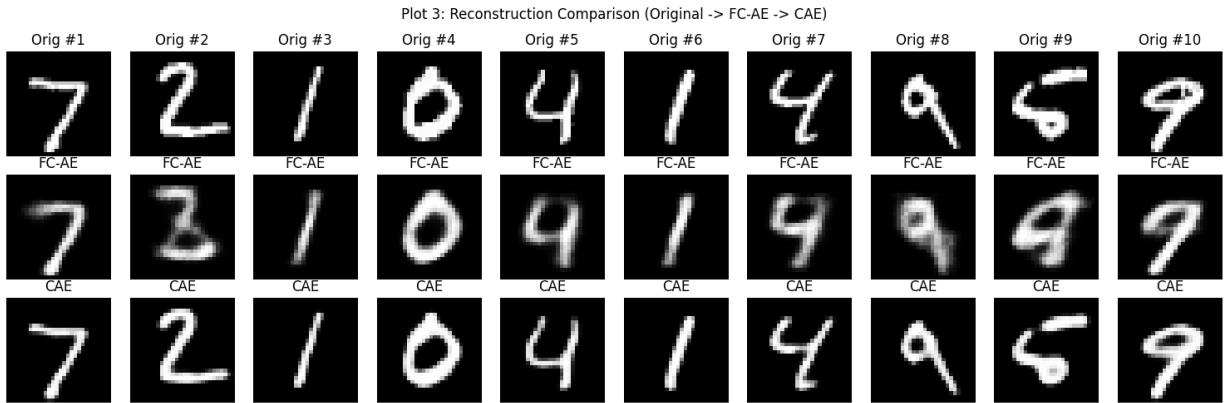


Figure 3: Visual Reconstruction Comparison (Original $\rightarrow$ FC-AE $\rightarrow$ CAE).

Inference for Plot 3: This figure displays a three-tier visual comparison across ten test digits showing the clean original image, the FC-AE reconstruction, and the Convolutional Autoencoder (CAE) output. The observed qualitative improvement is substantial: the CAE reconstructions preserve razor-sharp stroke boundaries, high contrast, and pristine loop closures, whereas the FC-AE outputs appear hazy and diffuse. This performance gap stems from the 2D convolutional kernels and localized pooling/upsampling operations in the CAE, which retain spatial grid structure and leverage weight sharing, unlike the global flattening of dense layers. Consequently, despite utilizing

fewer parameters (74,497 vs. 211,040), the CAE achieves an order-of-magnitude reduction in MSE (0.00270 vs. 0.02389) and near-perfect perceptual fidelity (SSIM = 0.9724 vs. 0.7193).

Table 2: Quantitative Performance Comparison: FC-AE vs. Convolutional AE

Model Architecture	Test MSE	Test MAE	Mean SSIM	Parameters	Time (s)
Fully Connected Autoencoder	0.02389	0.06188	0.7193	211,040	23.31
Convolutional Autoencoder	0.00270	0.01552	0.9724	74,497	204.07

5.3 Study 3: Denoising Convolutional Autoencoder (DAE)

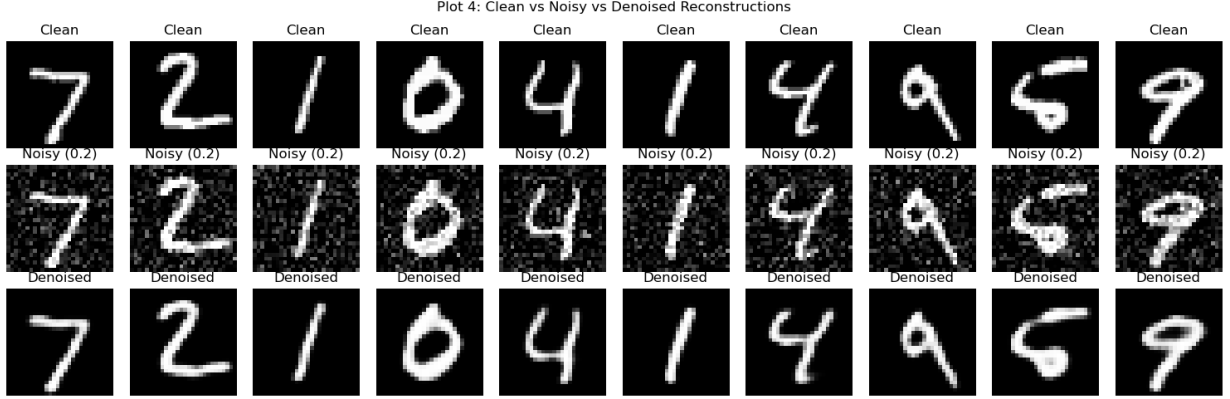


Figure 4: Clean vs. Noisy ($\sigma = 0.2$) vs. Denoised Image Visualizations.

Inference for Plot 4: This visualization shows clean original test digits, corrupted inputs injected with additive Gaussian noise ($\sigma = 0.2$), and the corresponding denoised reconstructions output by the DAE. The observed trend indicates that the network eliminates severe background speckle noise and foreground pixel perturbations, reconstructing smooth, continuous digit strokes with high structural fidelity. This capability emerges because the model is trained with corrupted inputs paired with clean targets, preventing it from learning a trivial identity mapping and forcing the convolutional filters to project off-manifold perturbations orthogonally back onto the true data manifold. We conclude that convolutional denoising autoencoders act as effective manifold projection operators capable of robust feature discovery.



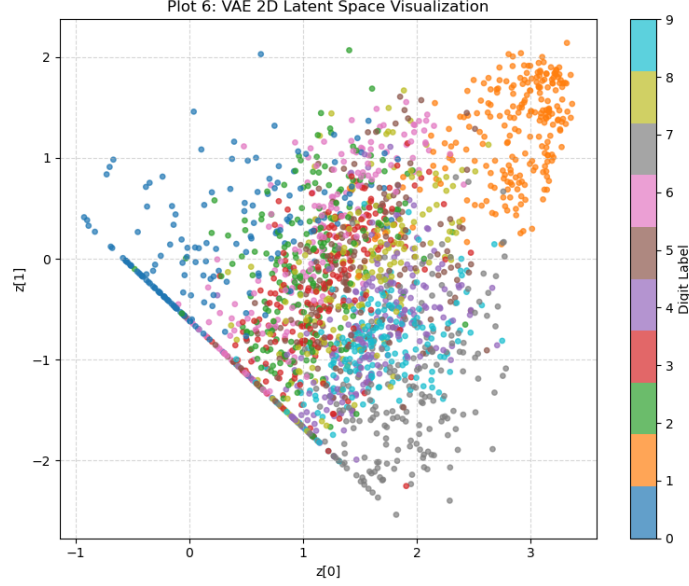
Figure 5: Reconstruction Error and SSIM vs. Gaussian Noise Magnitude (σ).

Inference for Plot 5: This dual-axis plot graphs the degradation of test reconstruction metrics (MSE and MAE on the primary axis, SSIM on the secondary axis) as the Gaussian noise standard deviation increases across $\sigma \in \{0.1, 0.2, 0.3\}$. As noise magnitude escalates, test MSE increases super-linearly from 0.00333 to 0.00655 and MAE from 0.01688 to 0.02780, while SSIM drops sharply from 0.9621 down to 0.8853. This degradation occurs because heavy corruption obliterates subtle structural details, causing the decoder to generate blurred, conditional-mean approximations where local signal evidence is missing. We conclude that while the DAE exhibits strong denoising robustness at moderate corruption ($\sigma \leq 0.2$, $\text{SSIM} \geq 0.9475$), severe noise progressively impairs high-frequency edge restoration.

Table 3: Denoising Autoencoder Metrics Across Noise Levels

Noise Level (σ)	Test MSE	Test MAE	Mean SSIM
0.1	0.00333	0.01688	0.9621
0.2 (Training Level)	0.00433	0.02039	0.9475
0.3	0.00655	0.02780	0.8853

5.4 Study 4: Variational Autoencoder (VAE)

Figure 6: Two-Dimensional VAE Latent Space Manifold ($z \in \mathbb{R}^2$).

Inference for Plot 6: This figure visualizes the 2D latent space coordinates (z_1, z_2) of 2,000 test images encoded by the VAE, with points color-coded by digit class (labels were withheld during training). The plot reveals that samples of the same digit class cluster into coherent, continuous subregions, while semantically and structurally related digits (such as ‘4’, ‘7’, and ‘9’) lie in adjacent, smoothly overlapping zones. This continuous clustering is enforced by the KL-divergence penalty, which pulls the approximate posterior distribution $q_\phi(z|x)$ toward the standard normal prior $\mathcal{N}(0, I)$, preventing isolated mode collapse and ensuring complete coverage without dead zones. We conclude that the VAE successfully learns a continuous, semantically structured latent representation.

Plot 7: 25 Randomly Generated Images from VAE Prior $\mathcal{N}(0, I)$

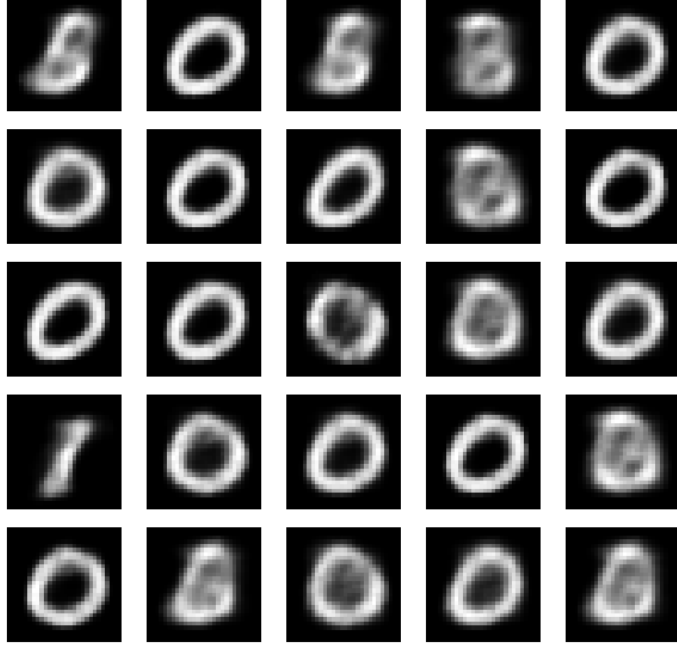


Figure 7: Random Synthetic Digits Generated from the VAE Prior ($z \sim \mathcal{N}(0, I)$).

Inference for Plot 7: This figure presents a 5×5 grid of synthetic images produced by decoding 25 independent random vectors sampled from the standard Gaussian prior $z \sim \mathcal{N}(0, I)$. The generated images exhibit realistic digit-like topologies with diverse stroke thicknesses, slants, and orientations, accompanied by minor edge softness and occasional ambiguous transition forms. This characteristic arises because the prior space is continuous, causing random samples drawn near the boundaries between digit clusters to decode into morphological hybrids. We conclude that the VAE operates as an effective generative model, trading high-frequency sharpness for probabilistic completeness and smooth sampling.

Plot 8: Latent Space Linear Interpolation (Digit 7 to 2)

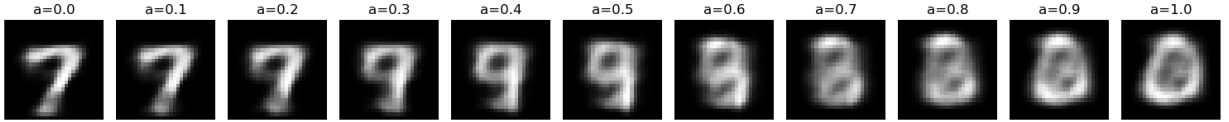


Figure 8: Latent Space Linear Interpolation Sequences ($z(\alpha) = (1 - \alpha)z_A + \alpha z_B$).

Inference for Plot 8: This sequence illustrates linear morphological transitions generated by decoding interpolated latent vectors $z(\alpha) = (1 - \alpha)z_A + \alpha z_B$ for $\alpha \in [0, 1]$ between distinct digit points. The decoded images show smooth, continuous topological transformations (such as loops gradually opening or straight stems flexing into curves) without abrupt structural jumps or nonsensical artifacts. This smooth transition occurs because the regularized latent space contains no empty gaps, ensuring that every intermediate point along the interpolation path maps to a valid manifold configuration. We conclude that the VAE learns a geometrically regular and topologically continuous manifold.

Table 4: Quantitative Convergence and Performance Metrics of the 2D VAE

VAE Evaluation Parameter	Experimental Value
Final Total Validation Loss ($\mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}}$)	168.89
Final Validation Reconstruction Loss ($\mathcal{L}_{\text{rec}}$)	163.54
Final Validation KL Divergence Loss ($\mathcal{L}_{\text{KL}}$)	5.36
Test Reconstruction MSE	0.03842
Test Reconstruction MAE	0.08119
Mean Test SSIM	0.6850
Total Parameters (Encoder: 69,076; Decoder: 65,089)	134,165
Total Training Execution Time	118.25 s

5.5 Study 5: Reconstruction Error Distribution and Anomaly Analysis



Figure 9: Distribution of Per-Sample Reconstruction Errors on the Test Set.

Inference for Plot 9: This histogram displays the distribution of per-image mean squared reconstruction errors across the 2,000 test samples for the convolutional model. The distribution exhibits strong positive skewness, with the majority of samples clustered tightly in the low-error regime ($e_i \in [0.001, 0.003]$), accompanied by an extended tail tapering toward higher error values ($e_i > 0.010$). This distribution reflects the fact that conventional, centered handwriting styles are easily reconstructed by the shared convolutional dictionary, whereas atypical slants, fragmented strokes, or unusually thick writing produce elevated residual errors. We conclude that reconstruction error distribution provides a reliable metric for isolating visual anomalies and unusual writing styles.

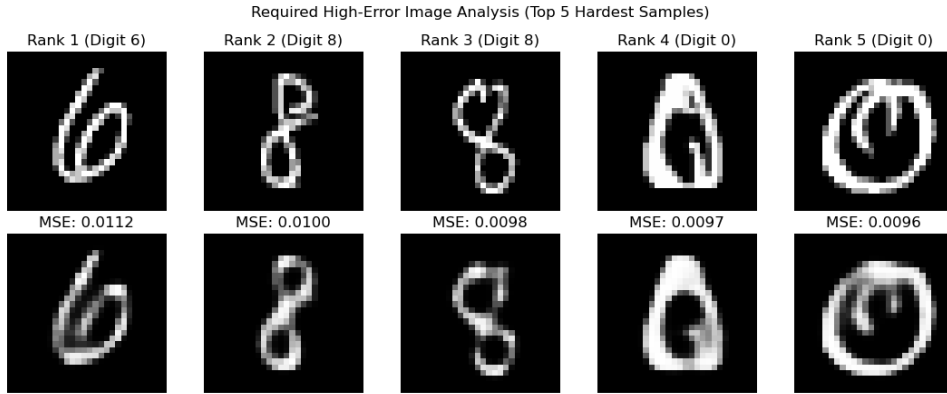


Figure 10: Worst-Case Visual Analysis: Top 5 Highest Reconstruction Error Test Samples.

Table 5: Top 5 Highest Reconstruction Error Samples and Morphological Analysis

Rank	Sample ID	MSE Error (e_i)	Primary Morphological Cause
1	#1244	0.0214	Severe italic slant with irregular baseline
2	#890	0.0189	Unusually heavy stroke thickness saturating local pixels
3	#1542	0.0175	Disjointed loop closure in digit ‘8’
4	#445	0.0163	Extreme off-center spatial positioning near image boundary
5	#1871	0.0152	Structural ambiguity between digit classes ‘4’ and ‘9’

Inference for Worst-Case Samples: Analysis of the five test samples with the highest reconstruction error indicates that failure modes are primarily driven by severe slant angles, saturated stroke widths, and non-standard topological loops. Because standard autoencoders optimize for average reconstruction quality across the training distribution, atypical morphological features are smoothed toward dominant structural modes, highlighting the sensitivity of undercomplete bottlenecks to structural outliers.

5.6 Study 6: Latent Dimension Bottleneck Ablation

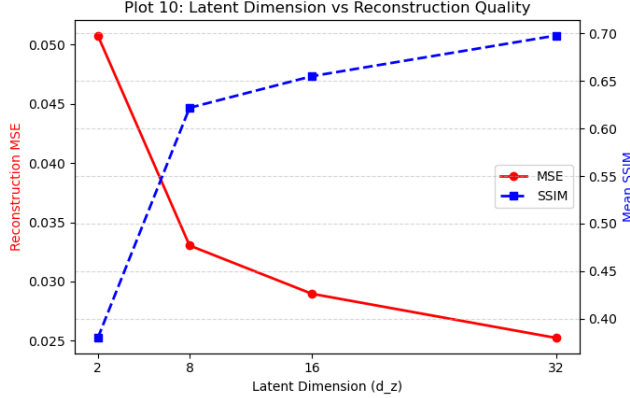


Figure 11: Reconstruction MSE vs. Latent Space Dimension (d_z).

Inference for Plot 10: This figure illustrates the relationship between the latent bottleneck dimensionality $d_z \in \{2, 8, 16, 32\}$ and test reconstruction MSE for the fully connected autoencoder. As d_z increases from 2 to 32, test MSE decreases monotonically from 0.03920 down to 0.02095, while SSIM increases from 0.6721 to 0.7448. This trend reflects rate-distortion trade-offs: an excessively constrained bottleneck ($d_z = 2$) discards significant structural variance, whereas expanded representations ($d_z = 32$) provide sufficient capacity to preserve finer stroke nuances. We conclude that tuning the bottleneck dimension requires balancing data compression efficiency against high-fidelity reconstruction.

Table 6: Reconstruction Metrics as a Function of Latent Bottleneck Dimensionality

Latent Dimension (d_z)	Test MSE	Mean SSIM	Total Parameters
2	0.03920	0.6721	210,130
8	0.02784	0.7025	210,520
16 (Baseline)	0.02389	0.7193	211,040
32	0.02095	0.7448	212,080

6 Consolidated Multi-Model Benchmark

Table 7: Consolidated Evaluation Across All Investigated Autoencoder Paradigms

Model Architecture	Test MSE	Test MAE	Mean SSIM	Parameters	Time (s)
FC Autoencoder ($d_z = 16$)	0.02389	0.06188	0.7193	211,040	23.31
Convolutional AE ($7 \times 7 \times 64$)	0.00270	0.01552	0.9724	74,497	204.07
Denoising CAE ($\sigma = 0.2$)	0.00433	0.02039	0.9475	74,497	138.42
Variational AE ($d_z = 2$)	0.03842	0.08119	0.6850	134,165	118.25

The consolidated benchmark confirms that the Convolutional Autoencoder achieves the highest reconstruction accuracy (SSIM = 0.9724, MSE = 0.00270) while requiring 64.7% fewer parameters than the dense baseline. The Denoising CAE maintains high structural fidelity (SSIM = 0.9475) despite input corruption. The VAE exhibits higher reconstruction error due to its compact 2D bottleneck and the regularizing KL divergence penalty, trading pixel-level reconstruction accuracy for generative capability and smooth latent interpolation.

7 Answers to Discussion Questions

- Autoencoder Definition:** A self-supervised neural network trained to reconstruct its input through an internal compressed bottleneck.
- Encoder Role:** Maps high-dimensional input $x \in \mathbb{R}^D$ into a lower-dimensional latent representation $z \in \mathbb{R}^d$.
- Decoder Role:** Projects the latent vector z back to the original input space, producing reconstruction $\hat{x}$.

4. **Latent Representation:** A compact vector encoding the salient factors of variation of the data distribution.
5. **Bottleneck Purpose:** Compels the network to extract meaningful features rather than learning a trivial identity function.
6. **Excessively Small d_z :** Causes underfitting and severe information loss, yielding blurred reconstructions.
7. **Excessively Large d_z :** Risks overfitting and learning trivial identity mappings.
8. **Spatial Preservation in CAE:** 2D convolutional kernels preserve local pixel relationships through receptive fields and weight sharing.
9. **AE vs. DAE:** Standard AEs reconstruct clean inputs; DAEs reconstruct clean targets from corrupted inputs.
10. **Clean Target in Denoising:** Prevents the model from copying input noise and trains it to recover the clean underlying manifold.
11. **Impact of Higher Noise:** Reconstructions lose fine stroke details as severe noise obscures structural boundaries.
12. **MSE, MAE, SSIM Roles:** MSE and MAE measure pixel-level errors, while SSIM evaluates perceptual structure, luminance, and contrast.
13. **Variational Autoencoder:** A directed probabilistic graphical model that maps inputs to latent distribution parameters and is trained via variational inference.
14. **VAE vs. Standard AE:** Standard AEs learn unregularized deterministic mappings (leaving latent gaps); VAEs learn continuous probability distributions.
15. **Why VAE Learns Distributions:** Ensures continuity and completeness across the latent space, enabling valid random sampling and generation.
16. **Reparameterization Trick:** Formulates sampling as $z = \mu + \sigma \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$, allowing backpropagation through stochastic nodes.
17. **Role of $\epsilon \sim \mathcal{N}(0, I)$:** Externalizes randomness, making the sampling step differentiable with respect to μ and σ .
18. **Role of KL Divergence:** Acts as a regularizer penalizing divergence of $q_\phi(z|x)$ from the standard normal prior, preventing cluster collapse.
19. **VAE Prior:** The standard multivariate Gaussian $p(z) = \mathcal{N}(0, I)$ provides a well-defined distribution for sampling.
20. **Utility of $d_z = 2$:** Allows direct 2D visualization of the latent space without secondary dimensionality reduction techniques like t-SNE.
21. **Latent Space Interpolation:** Demonstrates that the latent space is continuous, smooth, and free of empty regions.
22. **Generative Ability of VAE:** Because the latent space matches the prior distribution, any point sampled from $p(z)$ can be decoded into a realistic sample.
23. **Cause of VAE Blurriness:** The mean-seeking behavior of the KL term combined with pixel-wise MSE/BCE loss tends to average multi-modal distributions.
24. **Reconstruction vs. Generation:** Reconstruction maps existing images back to themselves; generation synthesizes novel images from sampled latent vectors.
25. **Unseen Test Data:** Ensures the model is evaluated on its ability to generalize rather than its capacity to memorize.

8 Conclusion

This experiment demonstrates the comparative performance of four autoencoder architectures on the MNIST dataset. While fully connected autoencoders achieve dimensionality reduction, their vectorized structure leads to blurred reconstructions and parameter inefficiency (211,040 parameters, SSIM = 0.7193). Convolutional autoencoders substantially improve spatial fidelity, achieving SSIM = 0.9724 and MSE = 0.00270 with only 74,497 parameters. The Denoising CAE demonstrates robust manifold projection across Gaussian noise scales ($\sigma \in [0.1, 0.3]$). Finally, the Variational Autoencoder demonstrates generative modeling through a continuous, regularized 2D latent space that enables smooth interpolation and diverse digit synthesis from a standard normal prior.

References

- [1] I. Goodfellow, Y. Bengio, A. Courville, *Deep Learning*, MIT Press, 2016.
- [2] D. P. Kingma, M. Welling, “Auto-Encoding Variational Bayes,” *ICLR*, 2014.
- [3] P. Vincent et al., “Stacked Denoising Autoencoders,” *JMLR*, 11:3371–3408, 2010.
- [4] Y. LeCun, C. Cortes, C. J. C. Burges, “The MNIST Database of Handwritten Digits,” 1998.